

Learned Incremental Nonlinear Dynamic Inversion for Quadrotors with and without Slung Payloads

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Abstract

The increasing complexity of multirotor applications demands flight controllers that can accurately account for all forces acting on the vehicle. Conventional controllers model most aerodynamic and dynamic effects but often neglect higher-order forces, as their accurate estimation is computationally expensive. Incremental Nonlinear Dynamic Inversion (INDI) offers an alternative by estimating residual forces from differences in sensor measurements; however, its reliance on specialized and often noisy sensors limits its applicability. Recent work has demonstrated that residual forces can be predicted using learning-based methods. In this paper, we show that a neural network can generate smooth approximations of INDI outputs without requiring specialized rotor RPM sensor inputs. We further propose a hybrid approach that integrates learning-based predictions with INDI and demonstrate both methods for multirotors and multirotors carrying slung payloads. Experimental results on trajectory tracking errors demonstrate that the specialized sensor measurements required by INDI can be eliminated by replacing the residual computation with a neural network.

Keywords: Learning-based control, Residual force estimation, Quadrotor, Slung payload.

1. Introduction

The rise in complexity of tasks to be executed by multirotors has necessitated the development of more accurate flight controllers that are able to model all forces acting on these robots. Traditional flight controllers (Lee et al., 2010) model a large part of these forces but are unable to model all of them. The remaining forces, called residual forces, can arise from diverse sources such as blade flapping, drag, or ground forces (Shi et al., 2020, 2022; Bauersfeld et al., 2021). Trying to compute residual forces directly can be computationally complex and intractable for real time control, especially on quadrotors with restricted hardware.

Recent work has shown that learning algorithms can lead to significant improvements in flight performance (Bauersfeld et al., 2021), even modeling the interaction forces between multiple quadrotors (Shi et al., 2020). By only learning the residual forces rather than the full dynamics of the quadrotor, the amount of training data needed is greatly reduced. In addition, learning models can help improve flight performance when sensor readings are noisy or lacking.

Incremental Nonlinear Dynamic Inversion (INDI) (Smeur et al., 2016) derives the residual forces from the mismatch between the nominal dynamic model and the real-time sensor feedback, using this residual to incrementally refine the control input. In the context of Unmanned Aerial Vehicles (UAVs), INDI typically relies on rotor RPM measurements to approximate the nominal model. Although INDI can significantly enhance flight performance, its effectiveness depends on the availability of specific sensor data and may still be affected by measurement noise.

针对带挂载和不带挂载的四旋翼的增量非线性动态反演学习

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摘要

多旋翼应用的日益复杂性要求飞行控制器能够准确考虑作用在飞行器上的所有力。传统控制器对大多数空气动力学和动态效应进行建模，但通常忽略高阶力，因为其精确估计计算成本高。增量非线性动态反演（INDI）提供了一种替代方法，通过传感器测量的差异来估计剩余力；然而，它依赖于专用且通常噪声较大的传感器，从而限制了其适用性。近期的研究表明，剩余力可以通过基于学习的方法进行预测。在本文中，我们展示了神经网络可以在不需要专用转子转速传感器输入的情况下生成INDI输出的平滑近似。我们进一步提出了一种将基于学习的预测与INDI整合的混合方法，并展示了这两种方法在多旋翼以及悬挂负载的多旋翼上的实验结果。实验证明了轨迹

关键词：基于学习的控制，残余力估计，四旋翼，悬挂负载。

1. 引言

多旋翼执行任务的复杂性增加，迫使开发能够模拟作用于这些机器人的所有力的更精确的飞行控制器。传统的飞行控制器（Lee 等，2010）模拟了这些力中的大部分，但无法模拟全部力。剩余力称为残余力，可能来自多种来源，例如桨叶摆动、阻力或地面力（Shi 等，2020，2022；Bauersfeld 等，2021）。直接计算残余力在计算上可能非常复杂，并且对于实时控制来说可能是无法解决的，尤其是在硬件受限的四旋翼上。

近期的研究表明，学习算法可以显著改善飞行性能（Bauersfeld 等，2021），甚至能够对多个四旋翼之间的相互作用力进行建模（Shi 等，2020）。通过仅学习残余力而不是四旋翼的完整动力学，所需的训练数据量大大减少。此外，在传感器读数存在噪声或不足的情况下，学习模型也有助于提高飞行性能。

增量非线性动态反演（INDI）（Smeur 等，2016）通过名义动态模型与实时传感器反馈之间的不匹配来推导剩余力，并利用该剩余力逐步优化控制输入。在无人机（UAV）的背景下，INDI 通常依赖转子转速测量来近似名义模型。尽管 INDI 可以显著提高飞行性能，但其有效性取决于特定传感器数据的可用性，并且仍可能受到测量噪声的影响。

One promising application of multirotors is payload transportation, particularly in time-sensitive fields such as medicine and rescue operations. When robots carry a payload attached by a cable (Tang et al., 2014; Wahba and Hönig, 2023), the dynamics of the multirotor become more complex due to the inclusion of the payload, which introduces additional forces on the system. These added dynamics can give rise to new types of unmodeled residual forces, thereby increasing the need for accurate estimators and making residual force prediction even more crucial.

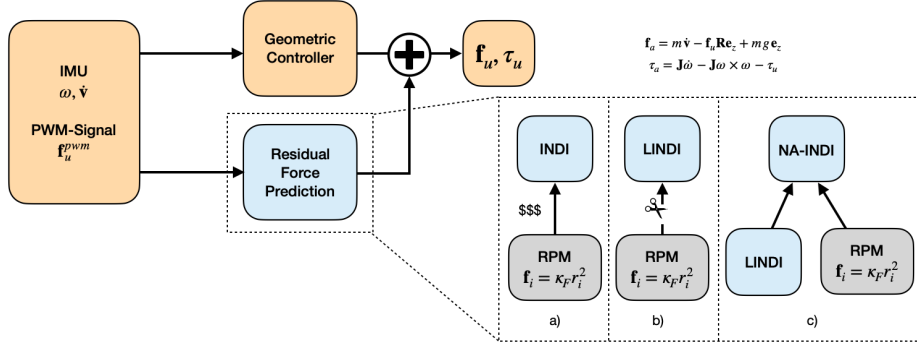


Figure 1: Overview of the three approaches for computing residual forces compared in our paper: a) Standard INDI relies on additional sensor measurements, which can be expensive and prone to noise. b) Learned INDI (LINDI) learns the outputs of INDI, eliminating the need for specialized sensors and yielding smoother predictions. c) Neural Augmented INDI (NA-INDI) fuses LINDI predictions with the original sensor data, resulting in higher-quality residual force estimates. LINDI and NA-INDI are formally defined in Section 4.

We propose Learned Incremental Nonlinear Dynamic Inversion (LINDI) a neural-network-based approach that achieves comparable flight performance to the INDI controller without requiring rotor RPM measurements. By replacing the incremental model with a learned component integrated into two traditional flight controllers (Lee et al., 2010; Yu et al., 2020), our approach removes the need for specialized RPM sensors. Quantitative results show that it matches INDI’s performance, and we further analyze a hybrid setup, Neural-Augmented Incremental Nonlinear Dynamic Inversion (NA-INDI), combining both INDI and neural network predictions. Figure 1 summarizes the three evaluated approaches.



Figure 2: Hardware used for experiments. We rely on a Bitcraze Crazyflie 2.1 with standard motors, Micro-SD-card extension, and custom RPM measurement board that uses IR-LEDs. These LEDs can also be tracked by a motion capture system. The payload is equipped with another active IR LED for tracking.

多旋翼的一项有前景的应用是运输载荷，尤其是在医疗和救援等时间敏感的领域。当机器人携带通过电缆连接的载荷时（Tang 等, 2014; Wahba 和 Hönig, 2023），由于载荷的加入，多旋翼的动力学变得更加复杂，这给系统引入了额外的力。这些额外的动力学可能产生新的未建模残余力，因此对精确估算器的需求增加，使残余力预测变得更加重要。

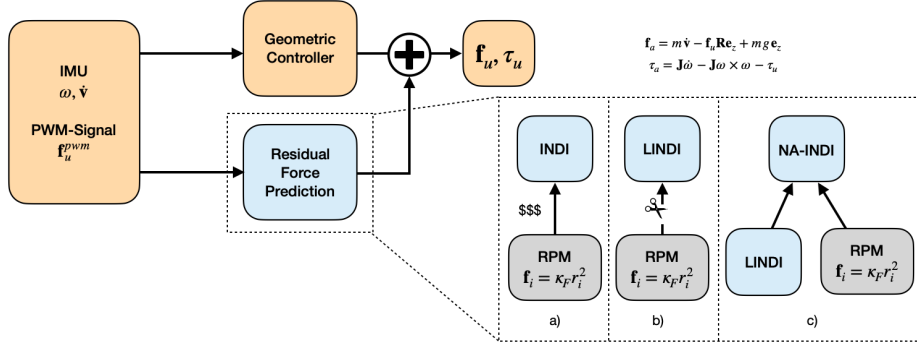


图1: 本文比较的三种残余力计算方法概述: a) 标准INDI依赖于额外的传感器测量, 这可能代价高且易受噪声影响。b) 学习型INDI (LINDI) 学习INDI的输出, 消除了对专用传感器的需求, 并产生更平滑的预测。c) 神经增强型INDI (NA-INDI) 将LINDI的预测与原始传感器数据融合, 从而获得更高质量的残余力估计。LINDI和NA-INDI在第4节中有正式定义。

我们提出了学习型增量非线性动态反演 (LINDI), 这是一种基于神经网络的方法, 能够在无需测量转子转速的情况下, 实现与INDI控制器相当的飞行性能。通过将增量模型替换为集成到两个传统飞行控制器中的学习组件 (Lee 等, 2010; Yu 等, 2020), 我们的方法消除了对专用转速传感器的需求。定量结果显示, 其性能与INDI相当, 我们还进一步分析了一种混合方案——神经增强增量非线性动态反演 (NA-INDI), 结合了INDI和神经网络预测。图1总结了三种评估方法。



图2: 实验使用的硬件。我们使用配备标准电机、Micro-SD卡扩展和自定义 RPM 测量板 (使用红外LED) 的 Bitcraze Crazyflie 2.1。这些LED也可以被运动捕捉系统跟踪。有效载荷配备了另一个主动红外LED用于跟踪。

2. Related Work

Several prior studies have highlighted the importance of INDI for achieving high tracking performance, particularly in high-speed flight. Specifically, incorporating the INDI component is essential for both MPC- and geometric-based controllers, with both approaches achieving comparable performance when provided with a valid reference trajectory (Sun et al., 2022). Performance can be further enhanced by modeling drag (Sun et al., 2022; Faessler et al., 2018) or motor delay (Tal and Karaman, 2021), though these factors have a comparatively smaller effect. More recently, INDI has also been shown to play a critical role in agile flight of multirotors carrying payloads (Sun et al., 2025).

An alternative to online estimation, as in INDI, is to learn a function that directly predicts the nonlinear residual dynamics. Bauersfeld et al. (2021) propose augmenting the controller with a neural network that predicts residual forces acting on a quadrotor. Their method adopts a non-Markovian assumption, using previous states as inputs rather than only the current state. The network outputs a six-dimensional vector representing the residual forces and torques along the x -, y -, and z -axes. Experimental results demonstrate that such learned residual models can significantly enhance flight performance. Similarly, Shi et al. (2022) investigate residual forces arising from inter-quadrotor interactions, particularly the downwash effect, where an upper quadrotor induces downward forces on a lower one. Their approach employs DeepSets to aggregate outputs from individual neural networks for each quadrotor, enabling the prediction of residual forces specifically along the z -axis. This method reduces the worst-case z -position error under downwash conditions by a factor of two to four.

Recent studies have explored integrating the INDI framework with learning-based models. Ignatyev and Tsourdos (2022) employ Gaussian Processes to correct sensor measurement inaccuracies in real time; their method updates the Gaussian Process online. In contrast, Zhang and Ran (2023) use meta-learning to continuously adapt residual predictions during flight, updating a neural network to estimate the INDI control effectiveness matrix. Our approach differs from previous work in that the model is trained offline and directly predicts INDI outputs, avoiding the instability that may occur when online models are not fully converged and removing the need for special sensor measurements. Additionally, unlike prior work, we validate our method through extensive real world evaluations on two different quadrotor modalities.

There is also prior work combining learning-based methods with slung-type payload transport in multirotors. Jin et al. (2024), for instance, employ a neural network to predict forces on a multirotor caused by an unknown payload. In contrast, our work assumes a known payload and focuses exclusively on residual dynamics.

The main contribution of our approach relative to previous works is that it eliminates the need for rotor RPM measurements while empirically demonstrating flight performance comparable to INDI, with smoother and less noisy residual force predictions.

3. Problem Description

We first introduce the dynamics that we consider in this work, followed by the control problems that we try to solve.

2. 相关工作

已有多项先前研究强调了INDI在实现高跟踪性能方面的重要性，尤其是在高速飞行中。具体而言，结合INDI组件对于基于MPC和基于几何的控制器都是必不可少的，两种方法在提供有效参考轨迹时能够实现相当的性能（Sun 等，2022）。通过建模阻力（Sun 等，2022；Faessler 等，2018）或电机延迟（Tal 和 Karaman，2021），性能可以进一步提升，尽管这些因素的影响相对较小。最近，研究还表明，INDI在承载负载的多旋翼无人机的敏捷飞行中也起着关键作用（Sun 等，2025）。

在线估计的另一种替代方法，如在INDI中，是学习一个可以直接预测非线性残差动力学的函数。Bauersfeld 等人（2021）提出通过神经网络增强控制器，该网络预测作用在四旋翼上的残差力。他们的方法采用非马尔可夫假设，使用先前状态作为输入，而不仅仅是当前状态。该网络输出一个六维向量，表示沿 x -、 y - 和 z -轴的残差力和力矩。实验结果表明，这类学习到的残差模型可以显著提高飞行性能。类似地，Shi 等人（2022）研究了由四旋翼之间相互作用产生的残差力，特别是下洗效应，即上层四旋翼对下层四旋翼产生向下的力。他们的方法使用 DeepSets 聚合每个四旋翼单独神经网络的输出，从而能够预测沿 z -轴的残差力。

最近的研究探索了将INDI框架与基于学习的模型结合。Ignatyev和Tsourdos（2022）使用高斯过程实时纠正传感器测量误差；他们的方法在线更新高斯过程。相比之下，Zhang和Ran（2023）使用元学习在飞行过程中持续调整残差预测，更新神经网络以估计INDI控制有效性矩阵。我们的方法与以往工作不同，因为我们的模型是在离线训练并直接预测INDI输出，从而避免了在线模型未完全收敛时可能出现的不稳定性，并且无需特殊的传感器测量。同时，不同于以往工作，我们通过对两种不同四旋翼模态进行大规模的真实环境评估来验证我们的方法。

在多旋翼的挂载式负载运输中，也有将基于学习的方法结合起来的先前工作。例如，Jin 等人（2024）使用神经网络来预测未知负载对多旋翼产生的力。相比之下，我们的工作假设负载已知，并专注于剩余动力学。

与以往工作相比，我们方法的主要贡献在于，它消除了对转子转速测量的需求，同时在经验上展示了可与INDI相媲美的飞行性能，并且剩余力的预测更加平滑且噪声更小。

3. 问题描述

我们首先介绍本工作中考虑的动态，然后介绍我们尝试解决的控制问题。

3.1. Multirotor Dynamics

Consider a multirotor modeled as a rigid floating base with state $\mathbf{x} = [\mathbf{p}, \mathbf{R}, \mathbf{v}, \boldsymbol{\omega}]^\top$. Here, $\mathbf{p}, \mathbf{v} \in \mathbb{R}^3$ represent the position and velocity in the world frame, $\mathbf{R} \in SO(3)$ represents the rotation matrix from body to world, and $\boldsymbol{\omega} \in \mathbb{R}^3$ is the angular velocity expressed in the body frame. The action $\mathbf{u} \in \mathbb{R}^4$ is defined as the angular velocities of the rotors, $\mathbf{u} = [r_1, r_2, r_3, r_4]^\top$. The dynamics are derived from Newton-Euler equations for rigid bodies as

$$\dot{\mathbf{p}} = \mathbf{v}, \quad m\dot{\mathbf{v}} = f_u \mathbf{R} \mathbf{e}_z - mg \mathbf{e}_z + \mathbf{f}_a, \quad (1)$$

$$\dot{\mathbf{R}} = \mathbf{R} \hat{\boldsymbol{\omega}}, \quad \mathbf{J} \dot{\boldsymbol{\omega}} = \mathbf{J} \boldsymbol{\omega} \times \boldsymbol{\omega} + \boldsymbol{\tau}_u + \boldsymbol{\tau}_a, \quad (2)$$

where m is the mass, $\mathbf{J}$ is the inertia matrix, g is the gravitational acceleration constant, $\mathbf{e}_z = [0, 0, 1]^\top$, f_u is the collective thrust created by the rotors, $\boldsymbol{\tau}_u$ is the torque vector created by the rotors, $\mathbf{f}_a, \boldsymbol{\tau}_a$ are unknown external forces and torques, respectively, and $\hat{\boldsymbol{\omega}}$ denotes the skew-symmetric matrix of $\boldsymbol{\omega}$ such that $\hat{\boldsymbol{\omega}} \mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}$ for any $\mathbf{v}$.

The relationship between the motor angular velocity and the generated total force and torque on the rigid body is

$$f_i = \kappa_F r_i^2, \quad (3a)$$

$$[f_u, \boldsymbol{\tau}_u]^\top = \mathbf{B}_0 [f_1, f_2, f_3, f_4]^\top, \quad (3b)$$

where κ_F is a propeller constant and $\mathbf{B}_0$ is a fixed and known actuation matrix.

3.2. Multirotor With Cable-Suspended Payload

Consider a slung type payload attached to the center of gravity of the multirotor. The multirotors state is extended to include the payload position $\mathbf{p}_p$ and velocity $\mathbf{v}_p$. The translational dynamics are extended with additional terms:

$$m\dot{\mathbf{v}} = f_u \mathbf{R} \mathbf{e}_z - mg \mathbf{e}_z + T \mathbf{q} + \mathbf{f}_a, \quad (4)$$

$$\dot{\mathbf{p}}_p = \mathbf{v}_p, \quad m_p \dot{\mathbf{v}}_p = -T \mathbf{q} - m_p g \mathbf{e}_z \quad (5)$$

where $\mathbf{q} = \frac{\mathbf{p}_p - \mathbf{p}}{\|\mathbf{p}_p - \mathbf{p}\|}$ indicates the direction of the cable, m_p is the mass of the payload, and T is the cable tension. From (4), the tension can be computed as

$$T = -m_p \mathbf{q}^\top (\dot{\mathbf{v}}_p + g \mathbf{e}_z). \quad (6)$$

3.3. Control Problem

Given the nominal models and feasible reference trajectories $\mathbf{x}_r(t)$, we are aiming to find controllers that can minimize the mean tracking error:

$$\operatorname{argmin}_{\pi} \frac{1}{D} \int_{t=0}^D d(\mathbf{x}_\pi(t), \mathbf{x}_r(t)) dt, \quad (7)$$

where $\pi(\mathbf{x}, \mathbf{x}_r) \mapsto \mathbf{u}$ is the control law that influences the state $\mathbf{x}_\pi(t)$, D the duration of the reference trajectory $\mathbf{x}_r(t)$, and d is a distance metric. For the multirotor case, we consider the positional tracking error, i.e., $d(\mathbf{x}_\pi, \mathbf{x}_r) = \|\mathbf{p}_\pi - \mathbf{p}_r\|_2$ and for the payload transport case the positional tracking error of the payload.

In this work, we augment classical controllers that ignore $\mathbf{f}_a$ and $\boldsymbol{\tau}_a$ with measured and/or predicted components to improve the tracking performance (7).

3.1. 多旋翼动力学

考虑一个多旋翼，其模型为具有状态 $\mathbf{x} = [\mathbf{p}, \mathbf{R}, \mathbf{v}, \boldsymbol{\omega}]^\top$ 的刚性浮动基座。在这里， $\mathbf{p}, \mathbf{v} \in \mathbb{R}^3$ 表示在世界坐标系中的位置和速度， $\mathbf{R} \in SO(3)$ 表示从机体到世界的旋转矩阵， $\boldsymbol{\omega} \in \mathbb{R}^3$ 表示在机体坐标系中表示的角速度。动作 $\mathbf{u} \in \mathbb{R}^4$ 定义为转子角速度， $\mathbf{u} = [r_1, r_2, r_3, r_4]^\top$ 。动力学由刚体的牛顿-欧拉方程导出为

$$\dot{\mathbf{p}} = \mathbf{v}, \quad m\dot{\mathbf{v}} = f_u \mathbf{R} \mathbf{e}_z - mg \mathbf{e}_z + \mathbf{f}_a, \quad (1)$$

$$\dot{\mathbf{R}} = \mathbf{R} \hat{\boldsymbol{\omega}}, \quad \mathbf{J} \dot{\boldsymbol{\omega}} = \mathbf{J} \boldsymbol{\omega} \times \boldsymbol{\omega} + \boldsymbol{\tau}_u + \boldsymbol{\tau}_a, \quad (2)$$

m 是质量， $\mathbf{J}$ 是惯性矩阵， g 是重力加速度常数， $\mathbf{e}_z = [0, 0, 1]^\top$ ， f_u 是由旋翼产生的总推力， $\boldsymbol{\tau}_u$ 是由旋翼产生的力矩向量， $\mathbf{f}_a, \boldsymbol{\tau}_a$ 分别是未知的外力和力矩， $\hat{\boldsymbol{\omega}}$ 表示 $\boldsymbol{\omega}$ 的斜对称矩阵，使得对于任意 $\mathbf{v}$ ，有 $\hat{\boldsymbol{\omega}} \mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}$ 。

电机角速度与刚体上产生的总力和力矩之间的关系是

$$f_i = \kappa_F r_i^2, \quad (3a)$$

$$[f_u, \boldsymbol{\tau}_u]^\top = \mathbf{B}_0 [f_1, f_2, f_3, f_4]^\top, \quad (3b)$$

其中 κ_F 是螺旋桨常数， $\mathbf{B}_0$ 是固定且已知的执行矩阵。

3.2. 带有缆索悬挂有效载荷的多旋翼

考虑一个挂载在多旋翼重心的悬挂型有效载荷。多旋翼的状态被扩展以包括有效载荷位置 $\mathbf{p}_p$ 和速度 $\mathbf{v}_p$ 。平移动力学通过附加项进行扩展：

$$m\dot{\mathbf{v}} = f_u \mathbf{R} \mathbf{e}_z - mg \mathbf{e}_z + T \mathbf{q} + \mathbf{f}_a, \quad (4)$$

$$\dot{\mathbf{p}}_p = \mathbf{v}_p, \quad m_p \dot{\mathbf{v}}_p = -T \mathbf{q} - m_p g \mathbf{e}_z \quad (5)$$

其中 $\mathbf{q} = \frac{\mathbf{p}_p - \mathbf{p}}{\|\mathbf{p}_p - \mathbf{p}\|}$ 表示电缆的方向， m_p 是有效载荷的质量， T 是电缆张力。从 (4) 式可以计算出张力为

$$T = -m_p \mathbf{q}^\top (\dot{\mathbf{v}}_p + g \mathbf{e}_z). \quad (6)$$

3.3. 控制问题

给定名义模型和可行参考轨迹 $\mathbf{x}_r(t)$ ，我们的目标是找到能够最小化平均跟踪误差的控制器：

$$\operatorname{argmin}_{\pi} \frac{1}{D} \int_{t=0}^D d(\mathbf{x}_\pi(t), \mathbf{x}_r(t)) dt, \quad (7)$$

其中 $\pi(\mathbf{x}, \mathbf{x}_r) \mapsto \mathbf{u}$ 是影响状态 $\mathbf{x}_\pi(t)$ 的控制律， D 是参考轨迹 $\mathbf{x}_r(t)$ 的持续时间， d 是一个距离度量。对于多旋翼情况，我们考虑位置跟踪误差，即 $d(\mathbf{x}_\pi, \mathbf{x}_r) = \|\mathbf{p}_\pi - \mathbf{p}_r\|_2$ ；对于负载运输情况，则考虑负载的位置信息跟踪误差。

在这项工作中，我们通过增加测量和/或预测的组件来增强忽略 $\mathbf{f}_a$ 和 $\boldsymbol{\tau}_a$ 的经典控制器，以提高跟踪性能 (7)。

4. Approach

This section introduces our two proposed methods. Learned Incremental Nonlinear Dynamic Inversion (LINDI) is a learning-based approach that eliminates the need for the additional sensor measurements required by INDI, while maintaining — or even improving — performance by learning from smoothed data. Neural-Augmented Incremental Nonlinear Dynamic Inversion (NA-INDI) combines the outputs of a neural network with INDI sensor measurements to achieve performance superior to both INDI and LINDI.

4.1. Geometric Control Laws

4.1.1. MULTIROTOR

An exponentially stable geometric controller for a multirotor computes the desired force and torques as follows (Lee et al., 2010):

$$f_u = (-\mathbf{K}_p \mathbf{e}_p - \mathbf{K}_v \mathbf{e}_v + m g \mathbf{e}_z + m \ddot{\mathbf{p}}_d - \mathbf{f}_a) \cdot \mathbf{R} \mathbf{e}_z, \quad (8a)$$

$$\begin{aligned} \boldsymbol{\tau}_u = & -\mathbf{K}_R \mathbf{e}_R - \mathbf{K}_\omega \mathbf{e}_\omega - \mathbf{J} \boldsymbol{\omega} \times \boldsymbol{\omega} \\ & - \mathbf{J} (\dot{\boldsymbol{\omega}} \mathbf{R}^\top \mathbf{R}_d \boldsymbol{\omega}_d - \mathbf{R}^\top \mathbf{R}_d \dot{\boldsymbol{\omega}}_d) - \boldsymbol{\tau}_a, \end{aligned} \quad (8b)$$

where $\mathbf{R}_d$, $\boldsymbol{\omega}_d$ and $\dot{\boldsymbol{\omega}}_d$ are desired references that can be computed using differential flatness; $\mathbf{e}_p$, $\mathbf{e}_v$, $\mathbf{e}_R$, $\mathbf{e}_\omega \in \mathbb{R}^3$ are errors with respect to this reference (mathematically defined in Lee et al. (2010)), and $\mathbf{K}_p$, $\mathbf{K}_v$, $\mathbf{K}_R$, $\mathbf{K}_\omega \in \mathbb{R}^{3 \times 3}$ are diagonal positive gain matrices.

The highlighted parts $\mathbf{f}_a$, $\boldsymbol{\tau}_a$ are new terms that need to be added to compensate for residual forces and torques.

4.1.2. MULTIROTOR WITH CABLE-SUSPENDED PAYLOAD

The previous control law can be extended to track a cable-suspended load (Yu et al., 2020). The controller is a cascaded design, where the first level tracks the position and velocity of the payload, the second level tracks the desired cable direction and its derivative, and the third level tracks the UAVs rotation and angular velocity. Eventually, f_u computes the control force generated by the multirotor to track the desired cable and payload motion, while attenuating the external residuals $\mathbf{f}_a$

$$f_u = (\mathbf{u}_0 - \mathbf{f}_a) \cdot \mathbf{R} \mathbf{e}_z, \quad (9)$$

where $\mathbf{u}_0$ is the nominal control force to track the cable as defined in Yu et al. (2020). Finally the rotational dynamics are identical to (8b), since the cable is assumed to be attached at the center of gravity and the mismatches between the nominal and real model are compensated using $\boldsymbol{\tau}_a$.

4.2. Incremental Nonlinear Dynamic Inversion (INDI)

The key idea of the Incremental Nonlinear Dynamic Inversion (INDI) is to estimate $\mathbf{f}_a$ and $\boldsymbol{\tau}_a$ in real-time using IMU and RPM sensor measurements. The INDI controller is implemented on the multirotor dynamics in Smeur et al. (2016); Tal and Karaman (2021). The dynamics of the multirotor-payload case differ from the standard case by the force applied by the cable. Therefore, re-arranging (2) and (4) yields

$$\mathbf{f}_a = m \dot{\mathbf{v}} - f_u \mathbf{R} \mathbf{e}_z + m g \mathbf{e}_z - T \mathbf{q}, \quad (10a)$$

$$\boldsymbol{\tau}_a = \mathbf{J} \dot{\boldsymbol{\omega}} - \mathbf{J} \boldsymbol{\omega} \times \boldsymbol{\omega} - \boldsymbol{\tau}_u. \quad (10b)$$

4. 方法

本节介绍了我们提出的两种方法。学习型增量非线性动态反演（LINDI）是一种基于学习的方法，它消除了 INDI 所需的额外传感器测量，同时通过从平滑数据中学习能够保持甚至提升性能。神经增强增量非线性动态反演（NA-INDI）将神经网络的输出与 INDI 传感器测量结合起来，实现了优于 INDI 和 LINDI 的性能。

4.1. 几何控制律

4.1.1. 多旋翼

一种指数稳定的多旋翼几何控制器按如下方式计算所需的力和力矩（Lee 等，2010）：

$$f_u = (-\mathbf{K}_p \mathbf{e}_p - \mathbf{K}_v \mathbf{e}_v + m g \mathbf{e}_z + m \ddot{\mathbf{p}}_d - \mathbf{f}_a) \cdot \mathbf{R} \mathbf{e}_z, \quad (8a)$$

$$\begin{aligned} \boldsymbol{\tau}_u = & -\mathbf{K}_R \mathbf{e}_R - \mathbf{K}_\omega \mathbf{e}_\omega - \mathbf{J} \boldsymbol{\omega} \times \boldsymbol{\omega} \\ & - \mathbf{J}(\dot{\boldsymbol{\omega}} \mathbf{R}^\top \mathbf{R}_d \boldsymbol{\omega}_d - \mathbf{R}^\top \mathbf{R}_d \dot{\boldsymbol{\omega}}_d) - \boldsymbol{\tau}_a, \end{aligned} \quad (8b)$$

其中 $\mathbf{R}_d$, $\boldsymbol{\omega}_d$ 和 $\dot{\boldsymbol{\omega}}_d$ 是可以使用微分平坦性计算的期望参考； $\mathbf{e}_p$, $\mathbf{e}_v$, $\mathbf{e}_R$, $\mathbf{e}_\omega \in \mathbb{R}^3$ 是相对于该参考的误差（在 Lee 等人 (2010) 中数学定义）， $\mathbf{K}_p$, $\mathbf{K}_v$, $\mathbf{K}_R$, $\mathbf{K}_\omega \in \mathbb{R}^{3 \times 3}$ 是对角正增益矩阵。

高亮部分 $\mathbf{f}_a$ 、 $\boldsymbol{\tau}_a$ 是需要添加的新术语，用于补偿剩余力和力矩。

4.1.2. 带有电缆悬挂有效载荷的多旋翼

先前的控制律可以扩展用于跟踪悬挂电缆的负载（Yu 等，2020）。该控制器采用级联设计，其中第一级跟踪负载的位置和速度，第二级跟踪所需的电缆方向及其导数，第三级跟踪无人机的旋转和角速度。最终， f_u 计算多旋翼产生的控制力以跟踪所需的电缆和负载运动，同时衰减外部残余 $\mathbf{f}_a$ 。

$$f_u = (\mathbf{u}_0 - \mathbf{f}_a) \cdot \mathbf{R} \mathbf{e}_z, \quad (9)$$

其中 $\mathbf{u}_0$ 是跟踪电缆的标称控制力，如 Yu 等人 (2020) 中定义。最后，旋转动力学与 (8b) 相同，因为假设电缆连接在重心处，并且使用 $\boldsymbol{\tau}_a$ 来补偿标称模型和实际模型之间的差异。

4.2. 增量非线性动态反演 (INDI)

增量非线性动态反演（INDI）的关键思想是利用 IMU 和 RPM 传感器测量实时估计 $\mathbf{f}_a$ 和 $\boldsymbol{\tau}_a$ 。INDI 控制器在 Smeur 等人 (2016) 以及 Tal 和 Karaman (2021) 中对多旋翼动力学进行了实现。多旋翼-有效载荷情况下的动力学与标准情况的区别在于电缆施加的力。因此，重新排列 (2) 和 (4) 得到

$$\mathbf{f}_a = m \dot{\mathbf{v}} - f_u \mathbf{R} \mathbf{e}_z + m g \mathbf{e}_z - T \mathbf{q}, \quad (10a)$$

$$\boldsymbol{\tau}_a = \mathbf{J} \dot{\boldsymbol{\omega}} - \mathbf{J} \boldsymbol{\omega} \times \boldsymbol{\omega} - \boldsymbol{\tau}_u. \quad (10b)$$

In INDI, f_u and τ_u are computed from RPM measurements by applying (3), $\dot{\mathbf{v}}$ is measured by the accelerometer in body frame and rotated to world frame, and $\boldsymbol{\omega}$ is measured by the gyroscope. Other values like $\dot{\boldsymbol{\omega}}$ and $\dot{\mathbf{v}}_p$ (needed through (6)) are estimated numerically, while $\mathbf{R}$ and $\mathbf{q}$ are computed by the state estimation. For practical purposes, it is important to filter the measured values, e.g., by using a butterworth filter (Tal and Karaman, 2021). The online estimated values can be used as additional feedforward terms in the controllers, as highlighted in (8) and (9).

Within the INDI framework, multirotor–payload dynamics can be modeled either by explicitly including payload and cable dynamics (Sreenath et al., 2013; Yu et al., 2020; Li et al., 2023), or by capturing the coupling via cable tension (Sun et al., 2025; Sreenath and Kumar, 2013). We adopt the latter formulation in (4) due to its practical suitability. The full formulation with explicit cable dynamics and the motivation for this choice are provided in the appendix.

4.3. Learned Incremental Nonlinear Dynamic Inversion (LINDI)

Using a dataset containing example trajectories with IMU data, RPM data, and state estimates, one can also compute $\mathbf{f}_a$ and $\boldsymbol{\tau}_a$ using (10). Here, noisy data such as $\dot{\mathbf{v}}$ and $\dot{\boldsymbol{\omega}}$ can be pre-processed with zero-delay filters such as spline fitting. The resulting values of $\mathbf{f}_a$ and $\boldsymbol{\tau}_a$ are the labels of a supervised learning problem.

In the single quadrotor case, we use a multi-layer perceptron (MLP) with 19 inputs, 6 outputs, 2 hidden layers with 24 dimensions each, and Leaky-ReLU activation. The input includes $\mathbf{v}$, $\dot{\mathbf{v}}$, $\boldsymbol{\omega}$, the first two columns of the rotation matrix $\mathbf{R}$, and the motor PWM signal (which is different from r_i as it cannot observe motor delays). The output is the residual force and torque $[\mathbf{f}_a, \boldsymbol{\tau}_a]^\top \in \mathbb{R}^6$. The inputs and outputs of the network are scaled using min-max normalization to the range $[-1, 1]$ to mitigate disparities in value magnitudes and enhance training stability.

For the payload network the input is expanded to include the payload velocity, acceleration and cable direction, resulting in 28 inputs. These quantities are included as they provide relevant information for estimating the residual forces acting on the payload. The architecture remains the same except for the hidden layers which are reduced to a dimension of 16. This is due to the residual forces in the payload scenario being more pronounced and making the training process easier allowing for a smaller network size.

For preprocessing, we fit splines composed of cubic polynomial segments that minimize the L_2 error with respect to the data points, rather than interpolating them exactly. These splines are applied to the collected INDI outputs, and the resulting smooth signals are used as training labels. The impact of training on smoothed data is illustrated in Figure 3. Because this smoothing is performed offline, it is more effective than the Butterworth filter used online in INDI.

4.4. Neural-Augmented Incremental Nonlinear Dynamic Inversion (NA-INDI)

We split the unmodeled dynamics in two parts $\mathbf{f}_a = \mathbf{f}_{a,NN} + \mathbf{f}_{a,INDI}$ and similar for $\boldsymbol{\tau}_a$, where $\mathbf{f}_{a,NN}$ and $\boldsymbol{\tau}_{a,NN}$ are learned functions that were trained using the steps described in 4.3. Then the INDI control law only needs to reason about the remaining mismatch in the dynamics.

5. Experiments

In order to quantify the performance of the different approaches, we perform real world flights on two unseen flight trajectories which are not present in the dataset used to train the neural network.

在 INDI 中, f_u 和 τ_u 通过应用公式 (3) 从转速测量值计算得出, $\dot{\mathbf{v}}$ 由机体坐标系下的加速度计测量并旋转到世界坐标系, ω 由陀螺仪测量。其他数值如通过公式 (6)) 需要的 ω 和 $\dot{\mathbf{v}}_p$ (则通过数值方法估计, 而 $\mathbf{R}$ 和 $\mathbf{q}$ 由状态估计计算得出。出于实际用途, 过滤测量值非常重要, 例如使用巴特沃斯滤波器 (Tal 和 Karaman, 2021)。在线估计的数值可以作为控制器中的附加前馈项使用, 如公式 (8) 和 (9) 中所强调的。

在 INDI 框架内, 多旋翼-载荷动力学可以通过显式包含载荷和索动力学 (Sreenath 等, 2013; Yu 等, 2020; Li 等, 2023) 来建模, 或者通过索力捕捉耦合关系 (Sun 等, 2025; Sreenath 和 Kumar, 2013)。我们采用公式 (4) 中的后者形式, 因为它在实际中更为适用。完整的显式索动力学公式及选择该方法的动机见附录。

4.3. 学习增量非线性动态反演 (L 印度)

使用包含带有 IMU 数据、RPM 数据和状态估计的示例轨迹的数据集, 也可以使用 (10) 计算 $\mathbf{f}_a$ 和 τ_a 。在这里, 像 $\dot{\mathbf{v}}$ 和 ω 这样的噪声数据可以通过零延迟滤波器 (例如样条拟合) 进行预处理。得到的 $\mathbf{f}_a$ 和 τ_a 的值是一个监督学习问题的标签。

在单旋翼机的情况下, 我们使用一个多层感知器 (MLP), 具有 19 个输入、6 个输出、2 个隐藏层, 每层 24 维, 并使用 Leaky-ReLU 激活。输入包括 $\mathbf{v}$ 、 $\dot{\mathbf{v}}$ 、 ω 、旋转矩阵的前两列 $\mathbf{R}$ 以及电机 PWM 信号 (与 r_i 不同, 因为它无法观测电机延迟)。输出是残余力和力矩 $[\mathbf{f}_a, \tau_a]^T \in \mathbb{R}^6$ 。网络的输入和输出使用最小-最大归一化缩放至范围 $[-1, 1]$, 以减轻数值大小差异并增强训练稳定性。

对于有效载荷网络, 输入被扩展以包括有效载荷的速度、加速度和电缆方向, 总共有 28 个输入。这些量被包括在内, 因为它们为估计作用在有效载荷上的残余力提供了相关信息。网络架构保持不变, 除了隐藏层被缩减到 16 维。这是因为在有效载荷场景中残余力更为显著, 使得训练过程更容易, 从而允许使用更小的网络规模。

在预处理过程中, 我们拟合了由三次多项式段组成的样条曲线, 这些曲线最小化了数据点的 L_2 误差, 而不是精确地插值。将这些样条曲线应用于收集的 INDI 输出, 得到的平滑信号用作训练标签。训练在平滑数据上的影响如图 3 所示。由于这种平滑是在离线进行的, 因此比 INDI 在线使用的巴特沃斯滤波器更有效。

4.4. 神经增强增量非线性动态反演 (NA-INDI)

我们将未建模的动力学分为两部分 $\mathbf{f}_a = \mathbf{f}_{a,NN} + \mathbf{f}_{a,INDI}$, τ_a 类似, 其中 $\mathbf{f}_{a,NN}$ 和 $\tau_{a,NN}$ 是通过 4.3 节中描述的步骤训练得到的学习函数。然后 INDI 控制律只需要考虑动力学中剩余的不匹配部分。

5. 实验

为了量化不同方法的性能, 我们在两个未在用于训练神经网络的数据集中出现的未知飞行轨迹上进行了实际飞行。

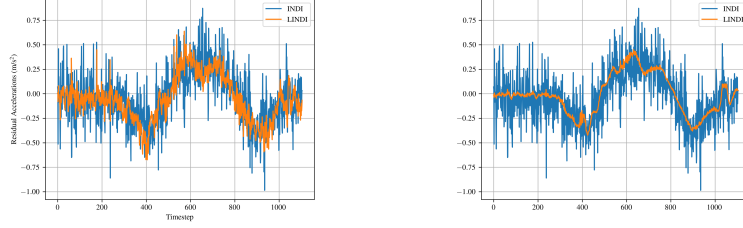


Figure 3: The figure shows the outputs of two different MLPs for the residual accelerations along the y-axis, compared with the original INDI predictions during a Figure8 flight path for a quadrotor carrying a payload. The left plot presents the predictions of an MLP trained on the raw INDI outputs, while the right plot shows the results from an MLP trained on spline-fitted INDI outputs. Although the standard MLP exhibits some inherent de-noising capability, it produces noticeably smoother predictions when trained on less noisy data, especially when using a smaller dataset for training.

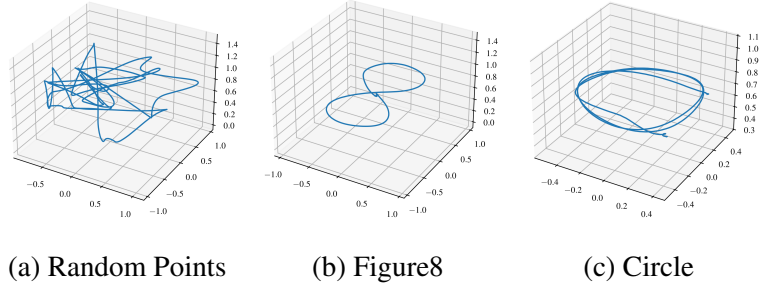


Figure 4: The first image (a) shows a sample trajectory used to generate training data for the neural network. Random points are sampled within a predefined bounding box to define a spline trajectory, which the quadrotor follows at a speed of up to 2 m/s. Images (b) and (c) show the trajectories used for testing the performance of the different controllers.

5.1. Physical Setup

We use the Bitcraze Crazyflie 2.1 quadrotors for our experiments, a small quadrotor with an arm length of 4.6 cm and weight of 34.7 g. It is operated by an STM32 microprocessor (168 MHz, 1 MB of flash, and 192 KB RAM). We equip the standard robot with two additions: a commercially available PCB to log data on a microSD card¹, and a custom PCB to measure RPM as can be seen in Figure 2. In order to accurately predict the position of the quadrotor, we use an OptiTrack motion capture system running at 100 Hz in a 7.5 m $\times$ 4 m $\times$ 2.75 m flight space. The custom PCB to measure RPMs uses infrared emitter/receivers and can be used as an active marker by the Optitrack motion capture system to estimate the robot’s pose. Rotors have reflective markers underneath the blade, which allow the IR receiver to count the rotor rotations. For communicating with the quadrotor we use Crazyswarm2 which is based on Crazyswarm (Preiss et al., 2017) but uses ROS 2 (Macenski et al., 2022).

1. <https://www.bitcraze.io/products/micro-sd-card-deck/>

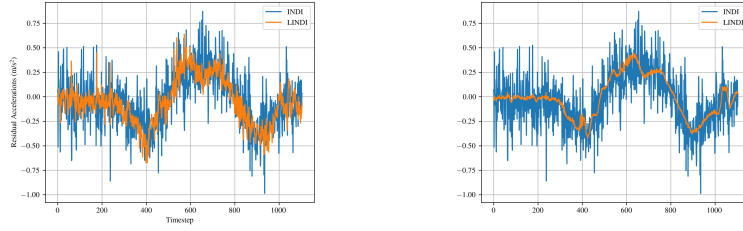


图3: 该图显示了两种不同多层感知器 (MLP) 对于沿y轴的残差加速度的输出, 并与四旋翼携带有效载荷在Figure8飞行路径期间的原始INDI预测进行比较。左图展示了在原始INDI输出上训练的MLP的预测结果, 而右图显示了在样条拟合INDI输出上训练的结果。尽管标准MLP具有一定的去噪能力, 但在使用噪声较少的数据进行训练时, 其预测结果会明显更平滑, 尤其是在使用较小的数据集进行训练时。

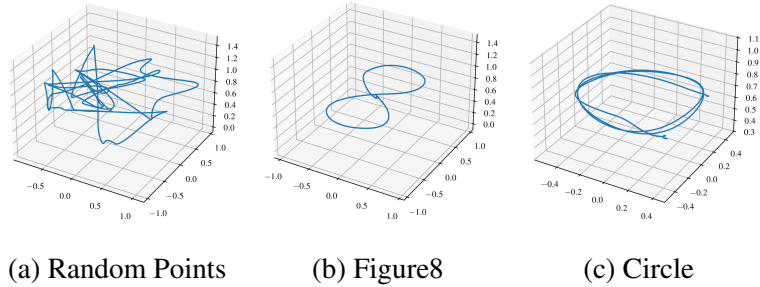


图4: 第一张图 (a) 显示了用于生成神经网络训练数据的示例轨迹。在预定义的边界框内随机采样点以定义样条轨迹, 四旋翼以最高2米/秒的速度沿该轨迹飞行。图 (b) 和 (c) 显示了用于测试不同控制器性能的轨迹。

5.1. 物理设置

我们在实验中使用 Bitcraze Crazyflie 2.1 四旋翼, 这是一种小型四旋翼, 臂长为 4.6 厘米, 重量为 34.7 克。它由 STM32 微处理器 (168 MHz, 1 MB 闪存, 192 KB RAM) 控制。我们在标准机器人上增加了两个组件: 一个商业可用的 PCB 用于在 microSD 卡上记录数据¹, 以及一个自定义 PCB 用于测量转速, 如图 2 所示。为了准确预测四旋翼的位置, 我们使用 OptiTrack 动作捕捉系统, 该系统在 7.5 m × 4 m × 2.75 m 的飞行空间内以 100 Hz 运行。用于测量转速的自定义 PCB 使用红外发射器/接收器, 并可以作为 OptiTrack 动作捕捉系统的主动标记, 用于估计机器人的姿态。旋翼叶片下方有反光标记, 使红外接收器能够计数旋翼的旋转次数。为了与四旋翼通信, 我们使用 Crazyswarm2, 它基于 Crazyswarm (Preiss 等, 2017), 但使用 ROS 2 (Macenski 等, 2022)。

1. <https://www.bitcraze.io/products/micro-sd-card-deck/>

For the payload scenario we use a 5 g weight attached at the center of the quadrotor by a string with 0.5 m length. The string is connected to the quadrotor by a magnet. The payload is also equipped with an infrared LED for estimating its position as can be seen in Figure 2. For simplicity, we keep the payload size and mass fixed, and refer the reader to prior work on adapting to varying payload parameters Jin et al. (2024).

5.2. Data Collection and Training

We train our neural networks on a set of trajectories generated by sampling random points within a predefined bounding box of dimensions $1.6 \times 1.6 \times 0.4, \text{m}^3$. These points are used to define spline trajectories, which the quadrotor follows at a randomly sampled velocity of up to 2 m/s in the no-payload case and up to 1.75 m/s in the payload case.

We collected a total of 64,996 timestamps, or around 12 minutes of flight data, for the no-payload scenario each one being a pair of state and residual force and torque, and 58,538 timestamps for the payload scenario.

We use the ADAM optimizer (Kingma and Ba, 2015) with an L_1 -loss and with an initial learning rate of $3 \cdot 10^{-1}$ while reducing the learning rate by a factor of 0.92 every 10 epochs to enhance training stability and convergence, training for a total of 128 epochs with a batch size of 512. The networks for the single UAV and the UAV with a payload are trained using the same procedure but with datasets corresponding to their respective scenarios. We found similar performance across hyperparameter settings in offline testing, so we omit a detailed ablation study. Our goal is to show that learning-based models can replace specialized sensor measurements, not to optimize a specific architecture.

We run all neural networks locally on the quadrotor hardware. For the MLP described in section 4.3, the inference time is 203 microseconds on the Crazyflie hardware.

5.3. Testing Scenarios

We test the performance of the controllers on two different predefined trajectories which are not present in the training set. The trajectories can be seen in Figure 4. We compute the mean tracking error (7) per trial and repeat over multiple trials (see captions). We do the same experiments for both the payload and no payload scenarios, with the only difference being the speed. For the scenario with no payload the maximum velocity was 2 m/s, while with the payload the maximum velocities were around 1.75 m/s.

5.4. Results

We compare our methods against the standard INDI controller and a geometric controller (Lee et al., 2010) with no residual force predictions.

5.4.1. IMPORTANCE OF RPM MEASUREMENTS FOR INDI

In order to prove the reliance on RPM measurements of INDI, we test the performance of an adjusted version of INDI which uses pulse width modulation (PWM) values instead of RPM to measure the total force output of the rotors. For this experiment we train LINDI and NA-INDI on the predictions made by INDI-PWM, with results shown in Table 1. PWM measurements are noisy and introduce delay-induced mismatch in thrust estimation.

对于载荷场景，我们使用一个通过0.5米长绳子固定在四旋翼中心的5克重物。绳子通过磁铁连接到四旋翼上。载荷还配备了一个红外LED，用于估算其位置，如图2所示。为简便起见，我们保持载荷的大小和质量固定，并请读者参考先前关于适应不同载荷参数的工作 Jin 等人（2024年）。

5.2. 数据收集与训练

我们在一组轨迹上训练我们的神经网络，这些轨迹是通过在预定义的边界框内随机采样点生成的，该边界框的尺寸为 $1.6 \times 1.6 \times 0.4, \text{m}^3$ 。这些点用于定义样条轨迹，四旋翼无人机按照随机采样的速度跟随这些轨迹，在无载荷情况下速度可达 2 m/s，而在有载荷情况下速度可达 1.75 m/s。

我们共收集了64,996个时间戳，约12分钟的飞行数据，用于无负载情景，每个时间戳都是状态与剩余力矩对；并收集了58,538个时间戳用于负载情景。

我们使用 ADAM 优化器 (Kingma 和 Ba, 2015)，配合 L_1 损失函数，以及初始学习率为 $3 \cdot 10^{-1}$ ，同时每 10 个训练周期将学习率降低 0.92 倍，以增强训练的稳定性 and 收敛性，总计训练 128 个周期，批量大小为 512。单无人机和带有效载荷的无人机的网络均使用相同的训练流程，但使用对应场景的数据集。我们在离线测试中发现超参数设置表现相似，因此省略了详细的消融研究。我们的目标是展示基于学习的模型能够替代专用传感器测量，而不是优化特定的架构。

我们在四旋翼硬件上本地运行所有神经网络。对于第4.3节中描述的多层感知器 (MLP)，在Crazyflie硬件上的推理时间为203微秒。

5.3. 测试场景

我们在两条训练集中不存在的不同预定义轨迹上测试控制器的性能。轨迹如图 4 所示。我们计算每次试验的平均跟踪误差 (7) 并在多次试验中重复（见图注）。对于有负载和无负载的情况，我们做了相同的实验，唯一的差别是速度。在无负载情况下，最大速度为 2 m/s，而有负载情况下，最大速度约为 1.75 m/s。

5.4. 结果

我们将我们的方法与标准的 INDI 控制器以及没有剩余力预测的几何控制器 (Lee 等, 2010) 进行比较。

5.4.1. RPM 测量的重要性

为了证明对 INDI 的转速 (RPM) 测量的依赖，我们测试了 INDI 的一个调整版本的性能，该版本使用脉宽调制 (PWM) 值而不是转速来测量转子的总推力输出。在这个实验中，我们训练了 LINDI 和 NA-INDI 以预测 INDI-PWM 的输出，结果如表 1 所示。PWM 测量存在噪声，并且在推力估计中引入了由延迟引起的不匹配。

Table 1: Comparison of average deviation from desired flight path (centimeters) of the quadrotor with no payload using INDI based on thrust from the PWM signal instead of RPM. The performance of each method was averaged over ten trials.

	Circle	Figure8
Lee	7.52 ± 1.22	7.29 ± 0.31
INDI-PWM	18.27 ± 0.97	18.89 ± 1.82
LINDI-PWM	15.33 ± 1.90	14.53 ± 1.75
NA-INDI-PWM	14.31 ± 1.91	16.71 ± 2.80

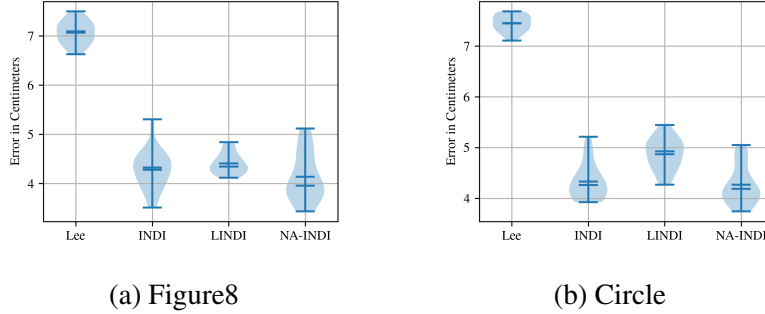


Figure 5: Performance comparison between all four methods on the quadrotor with no payload for (a) the Figure8 flight trajectory and (b) the circle trajectory. Each violin plot has twenty datapoints: the mean tracking error (centimeters) of the multirotor position (7) for each trial. Mean values for Figure8: Lee: 7.07, INDI: 4.28, LINDI: 4.41, NA-INDI: **4.14**. Mean values for Circle: Lee: 7.45, INDI: 4.33, LINDI: 4.87, NA-INDI: **4.26**.

The results show that using LINDI and NA-INDI to learn the residual force calculations from the PWM-based INDI greatly improves flight performance compared to the PWM-based INDI but still underperformed when compared to the basic Lee controller with no residual prediction. This highlights INDI's dependence on accurate sensor measurements. In addition, we observe that the RPM measurements are still required for training our learning methods. The improvement over INDI-PWM with LINDI-PWM and NA-INDI-PWM indicate that the denoising capabilities of LINDI can potentially improve the performance of the standard INDI.

5.4.2. MULTIROTOR WITH NO PAYLOAD

Figure 5 shows that residual prediction can reduce the tracking error to the desired flight trajectory by around 40%. As can be seen from the results, LINDI leads to similar performance to INDI without the use of the RPM measurements, however, for the circular trajectory, we observe a slight drop in performance, likely because the trajectory is out of distribution relative to the training data. The results also show that the combination of sensor measurements plus learned residuals (NA-INDI) leads to the best performance, although with a small margin. This results from the noise in the RPM measurements being easier to filter, as the INDI component only needs to account for a smaller portion of the residuals, with the remaining effects handled by the LINDI component.

表 1: 使用基于 PWM 信号推力而非 RPM 的 INDI 控制的四旋翼在无载荷情况下偏离期望飞行路径的平均偏差（厘米）比较。每种方法的性能均取十次试验的平均值。

	Circle	Figure8
Lee	7.52 ± 1.22	7.29 ± 0.31
INDI-PWM	18.27 ± 0.97	18.89 ± 1.82
LINDI-PWM	15.33 ± 1.90	14.53 ± 1.75
NA-INDI-PWM	14.31 ± 1.91	16.71 ± 2.80

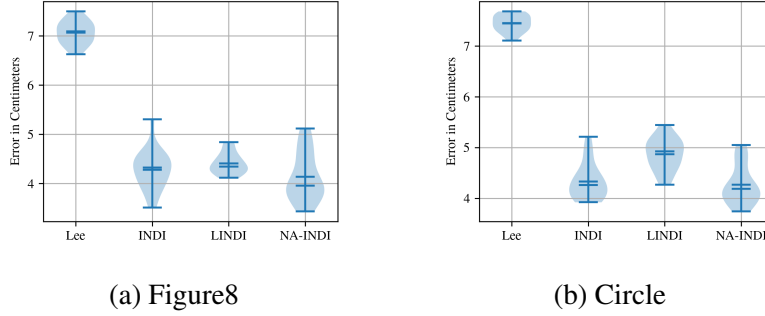


图5: 在无人机无载荷情况下, 四种方法在 (a) 8字形飞行轨迹和 (b) 圆形轨迹上的性能比较。每个小提琴图包含二十个数据点: 每次试验中多旋翼位置 (7) 的平均跟踪误差 (厘米)。8字形平均值: Lee: 7.07, INDI: 4.28, LINDI: 4.41, NA-INDI: 4.14。圆形平均值: Lee: 7.45, INDI: 4.33, LINDI: 4.87, NA-INDI: 4.26。

结果表明, 与基于PWM的INDI相比, 使用LINDI和NA-INDI从基于PWM的INDI中学习剩余力计算可以大大提高飞行性能, 但与不进行剩余预测的基础Lee控制器相比仍然表现不足。这突显了INDI对精确传感器测量的依赖。此外, 我们观察到训练我们的学习方法仍然需要RPM测量。与LINDI-PWM和NA-INDI-PWM相比, INDI-PWM的性能提升表明, LINDI的去噪能力有可能提高标准INDI的性能。

5.4.2. 无有效载荷的多旋翼

图5显示, 残差预测可以将跟踪误差减少到期望飞行轨迹约40%。从结果可以看出, LINDI在不使用RPM测量的情况下, 其性能与INDI相似。然而, 对于圆形轨迹, 我们观察到性能略有下降, 这可能是因为轨迹相对于训练数据的分布有所偏离。结果还显示, 传感器测量与学习残差的结合 (NA-INDI) 能够实现最佳性能, 尽管差距不大。这是因为RPM测量中的噪声更容易被滤除, INDI组件只需要处理残差的一小部分, 其余的影响由LINDI组件处理。

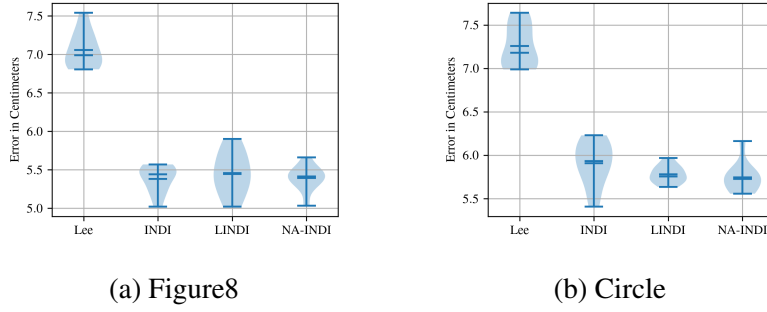


Figure 6: Performance comparison between all four methods on the quadrotor carrying a payload for (a) the Figure8 flight trajectory and (b) the circle trajectory. Each violin plot has ten datapoints: the mean tracking error (centimeters) of the payload position (7) for each trial. Mean values for Figure8: Lee: 7.06, INDI: **5.38**, LINDI: 5.46, NA-INDI: 5.39. Mean values for Circle: Lee: 7.26, INDI: 5.91, LINDI: 5.78, NA-INDI: **5.74**.

5.4.3. MULTIROTOR WITH PAYLOAD

In Figure 6 we can see similar results to the single quadrotor experiments. Once again LINDI is able to match the performance of INDI without the need of RPM measurements, even with the more complex dynamics of the payload. NA-INDI shows the potential to outperform INDI, but still remains similar in performance.

6. Conclusion

We introduce LINDI, a novel approach for learning residual predictions which trains a neural network on smooth data based on spline fitting. The results demonstrate that INDI can be effectively replaced by a neural network, which can output smoother residual force predictions without requiring special sensor measurements. While the proposed method requires a quadrotor equipped with RPM measurement sensors to collect the training data, once the neural network has been trained, it can be deployed on any number of quadrotors with similar specifications. This can significantly reduce the cost of maintaining a large fleet of drones with accurate flight controllers.

This work establishes a foundation for extending learning-based residual modeling toward more realistic scenarios involving external aerodynamic disturbances. Exploring how LINDI performs in the presence of such external effects offers a promising avenue for future research and could enable robust, disturbance-aware residual prediction for real-world operations.

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References

- L. Bauersfeld, E. Kaufmann, P. Foehn, S. Sun, and D. Scaramuzza. Neurobem: Hybrid aerodynamic quadrotor model. In *Robotics: Science and Systems XVII*. Robotics: Science and Systems

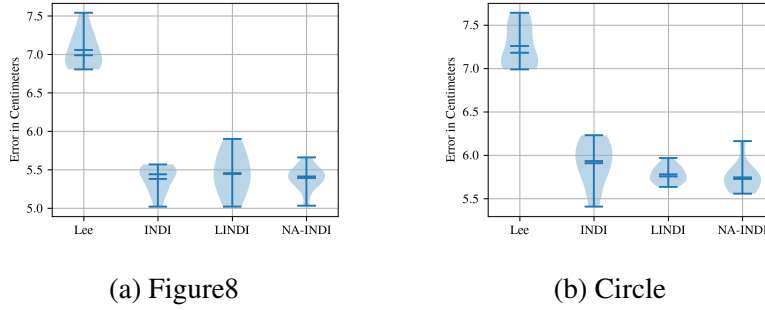


图6: 四种方法在载有效载荷的四旋翼上性能比较: (a) Figure8 飞行轨迹, (b) 圆形轨迹。每个小提琴图有十个数据点: 每次试验中有效载荷位置 (7) 的平均跟踪误差 (厘米)。Figure8 的平均值: Lee: 7.06, INDI: 5.38, LINDI: 5.46, NA-INDI: 5.39。圆形轨迹的平均值: Lee: 7.26, INDI: 5.91, LINDI: 5.78, NA-INDI: 5.74。

5.4.3. 带有效载荷的多旋翼

在图6中, 我们可以看到与单个四旋翼实验相似的结果。LINDI再次能够在不需要RPM测量的情况下匹配INDI的性能, 即使在载荷更复杂的动力学下。NA-INDI显示出超越INDI的潜力, 但性能仍然相似。

6. 结论

我们介绍了LINDI, 一种用于学习残差预测的新方法, 该方法通过基于样条拟合的平滑数据训练神经网络。结果表明, INDI可以被神经网络有效替代, 神经网络可以输出更平滑的残差力预测, 而无需特殊的传感器测量。虽然所提出的方法需要配备RPM测量传感器的四旋翼来收集训练数据, 但一旦神经网络训练完成, 它可以部署在具有类似规格的任何数量的四旋翼上。这可以显著降低维护拥有精确飞行控制器的大型无人机队的成本。

这项工作为将基于学习的残差建模扩展到涉及外部气动扰动的更现实场景奠定了基础。探索 LINDI 在存在此类外部影响时的表现, 为未来研究提供了有前景的方向, 并可能实现对实际操作中稳健的、感知干扰的残差预测。

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参考文献

L. Bauersfeld, E. Kaufmann, P. Foehn, S. Sun 和 D. Scaramuzza. Neurobem: 混合气动四旋翼模型。在 *Robotics: Science and Systems XVII* 中。机器人学: 科学与系统

- Foundation, 2021.
- Matthias Faessler, Antonio Franchi, and Davide Scaramuzza. Differential flatness of quadrotor dynamics subject to rotor drag for accurate tracking of high-speed trajectories. *IEEE Robotics and Automation Letters*, 3(2):620–626, 2018. doi: 10.1109/LRA.2017.2776353.
- Dmitry Ignatyev and Antonios Tsourdos. Incremental nonlinear dynamic inversion with sparse online gaussian processes adaptation for partially unknown systems. In *Mediterranean Conference on Control and Automation (MED)*, pages 233–238, 2022. doi: 10.1109/MED54222.2022.9837175.
- Ao Jin, Chenhao Li, Qinyi Wang, Ya Liu, Panfeng Huang, and Fan Zhang. Neural predictor for flight control with payload. *arXiv preprint arXiv:2410.15946*, 2024. URL <https://arxiv.org/abs/2410.15946>.
- D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*, 2015.
- T. Lee, M. Leok, and N. H. McClamroch. Geometric tracking control of a quadrotor uav on $se(3)$. In *IEEE Conference on Decision and Control (CDC)*, 2010.
- Guanrui Li, Xinyang Liu, and Giuseppe Loianno. Rotortm: A flexible simulator for aerial transportation and manipulation. *IEEE Transactions on Robotics*, 40:831–850, 2023.
- S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall. Robot operating system 2: Design, architecture, and uses in the wild. *Science Robotics*, 7(66), 2022.
- J. A. Preiss, W. Hönig, G. S. Sukhatme, and N. Ayanian. Crazyswarm: A large nano-quadcopter swarm. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3299–3304, 2017.
- Guanya Shi, Wolfgang Hönig, Yisong Yue, and Soon-Jo Chung. Neural-swarm: Decentralized close-proximity multirotor control using learned interactions. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.
- Guanya Shi, Wolfgang Hönig, Xichen Shi, Yisong Yue, and Soon-Jo Chung. Neural-swarm2: Planning and control of heterogeneous multirotor swarms using learned interactions. *IEEE Transactions on Robotics*, 38(2):1063–1079, 2022. doi: 10.1109/TRO.2021.3098436.
- Ewoud J. J. Smeur, Qiping Chu, and Guido C. H. E. Croon. Adaptive incremental nonlinear dynamic inversion for attitude control of micro air vehicles. *American Institute of Aeronautics and Astronautics*, 39(3):450–461, 2016. doi: 10.2514/1.G001490.
- Koushil Sreenath and Vijay R. Kumar. Dynamics, control and planning for cooperative manipulation of payloads suspended by cables from multiple quadrotor robots. In *Robotics: Science and Systems*, 2013.
- Koushil Sreenath, Taeyoung Lee, and Vijay Kumar. Geometric control and differential flatness of a quadrotor uav with a cable-suspended load. In *IEEE conference on decision and control*, pages 2269–2274, 2013.

Foundation, 2021年。

Matthias Faessler, Antonio Franchi 和 Davide Scaramuzza。受转子阻力影响的四旋翼动力学的微分平坦性, 用于高速度轨迹的精确跟踪。 *IEEE Robotics and Automation Letters*, 3(2):620–626, 2018. doi: 10.1109/LRA.2017.2776353.

德米特里·伊格纳季耶夫 (Dmitry Ignatyev) 和安东尼奥斯·特苏尔多斯 (Antonios Tsourdos)。用于部分未知系统的具有稀疏在线高斯过程适应的增量非线性动态反演。在 *Mediterranean Conference on Control and Automation (MED)*, 第 233–238 页, 2022 年。doi: 10.1109/MED54222.2022.9837175。

Ao Jin, 李晨浩, 王琴怡, 刘雅, 黄攀峰, 和 张凡. 具有有效载荷的飞行控制神经预测器。 *arXiv preprint arXiv:2410.15946*, 2024. URL <https://arxiv.org/abs/2410.15946>.

D. P. Kingma 和 J. Ba. Adam: 一种随机优化方法。在 *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*, 2015 年。

T. Lee, M. Leok 和 N. H. McClamroch。四旋翼无人机在 $se(3)$ 上的几何跟踪控制。发表于 *IEEE Conference on Decision and Control (CDC)*, 2010。

李冠锐, 刘新阳, 和 Giuseppe Loianno. Rotortm: 一个用于空中运输和操作的灵活模拟器。 *IEEE Transactions on Robotics*, 40:831–850, 2023.

S. Macenski, T. Foote, B. Gerkey, C. Lalancette, 和 W. Woodall。《机器人操作系统 2: 设计、架构及实际应用》。 *Science Robotics*, 7(66), 2022。

J. A. Preiss, W. Hönig, G. S. Sukhatme 和 N. Ayanian. CrazySwarm: 一个大型纳米四旋翼无人机群。在 *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 第 3299–3304 页, 2017 年。

Guanya Shi, Wolfgang Hönig, Yisong Yue 和 Soon-Jo Chung. Neural-swarm: 使用学习到的交互进行去中心化近距离多旋翼控制。发表于 *IEEE International Conference on Robotics and Automation (ICRA)*, 2020。

Guanya Shi, Wolfgang Hönig, Xichen Shi, Yisong Yue, 和 Soon-Jo Chung. Neural-swarm2: 使用学习到的交互进行异构多旋翼群体的规划与控制。 *IEEE Transactions on Robotics*, 38(2):1063–1079, 2022. doi: 10.1109/TRO.2021.3098436.

Ewoud J. J. Smeur, Qiping Chu 和 Guido C. H. E. Croon。用于微型飞行器姿态控制的自适应增量非线性动态反演。《v20》《v21》, 39(3):450–461, 2016. doi: 10.2514/1.G001490。

Koushil Sreenath 和 Vijay R. Kumar。多四旋翼机器人通过绳索悬挂的有效载荷的协作操作的动力学、控制与规划。在 *Robotics: Science and Systems*, 2013 年。

Koushil Sreenath, Taeyoung Lee, 和 Vijay Kumar。带有绳索悬挂负载的四旋翼无人机的几何控制与微分平坦性。发表于 *IEEE conference on decision and control*, 第 2269–2274 页, 2013 年。

- Sihao Sun, Angel Romero, Philipp Foehn, Elia Kaufmann, and Davide Scaramuzza. A comparative study of nonlinear mpc and differential-flatness-based control for quadrotor agile flight. *IEEE Transactions on Robotics (T-RO)*, 2022. Preprint available at https://rpg.ifi.uzh.ch/docs/TR022_Sun.pdf.
- Sihao Sun, Xuerui Wang, Dario Sanalitra, Antonio Franchi, Marco Tognon, and Javier Alonso-Mora. Agile and cooperative aerial manipulation of a cable-suspended load. *Science Robotics*, 10(107):eadu8015, 2025.
- Ezra Tal and Sertac Karaman. Accurate tracking of aggressive quadrotor trajectories using incremental nonlinear dynamic inversion and differential flatness. *IEEE Transactions on Control Systems Technology*, 29(3):1203–1218, 2021. doi: 10.1109/TCST.2020.3001117.
- Sarah Tang, Koushil Sreenath, and Vijay Kumar. Aggressive maneuvering of a quadrotor with a cable-suspended payload. In *Robotics: Science and Systems, Workshop on Women in Robotics*, pages 1–20, 2014.
- Khaled Wahba and Wolfgang Hönig. Efficient optimization-based cable force allocation for geometric control of a multirotor team transporting a payload. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1234–1240, 2023. doi: 10.1109/ICRA48506.2023.1234567.
- Beomyeol Yu, Kanishke Gamagedara, Seungkeun Kim, Taeyoung Lee, and Jinyoung Suk. Geometric control and experimental validation for a quadrotor uav transporting a payload. In *IEEE Conference on Decision and Control (CDC)*, pages 201–207, 2020.
- Xinyue Zhang and Maopeng Ran. Meta-learning-based incremental nonlinear dynamic inversion control for quadrotors with disturbances. *Applied Sciences*, 13(21):11844, 2023. doi: 10.3390/app132111844. Special Issue: Intelligent Unmanned System Technology and Application.

Sihao Sun, Angel Romero, Philipp Foehn, Elia Kaufmann 和 Davide Scaramuzza. 四旋翼机敏捷飞行的非线性 MPC 与基于微分平坦性控制的比较研究。 *IEEE Transactions on Robotics (T-RO)*, 2022. 预印本可在 https://rpg.ifi.uzh.ch/docs/TRO22_Sun.pdf 获取。 Sihao Sun, Xuerui Wang, Dario Sanalistro, Antonio Franchi, Marco Tognon 和 Javier Alonso-Mora. 缆索悬挂负载的灵活合作航空操作。 *Science Robotics*, 10(107):eadu8015, 2025. Ezra Tal 和 Sertac Karaman. 使用增量非线性动态反演和微分平坦性精确跟踪激进的四旋翼轨迹。 *IEEE Transactions on Control Systems Technology*, 29(3):1203–1218, 2021. doi: 10.1109/TCS T.2020.3001117. Sarah Tang, Koushil Sreenath 和 Vijay Kumar. 携带缆索悬挂负载的四旋翼激进机动。 在 *Robotics: Science and Systems, Workshop on Women in Robotics*, 页码 1–20, 2014. Khaled Wahba 和 Wolfgang Hönig. 用于多旋翼团队运输负载的几何控制的高效优化缆索力分配。 在 *2023 IEEE International Conference on Robotics and Automation (ICRA)*

Appendix A. Cable-Suspended Payload Dynamics

The dynamics of a single quadrotor transporting a point-mass payload via a cable modeled as a rigid rod are governed by the following equations. According to [Tang et al. \(2014\)](#); [Sreenath et al. \(2013\)](#); [Yu et al. \(2020\)](#); [Li et al. \(2023\)](#), the coupled quadrotor–payload dynamics can be expressed from Hamilton’s principle as

$$\begin{aligned} m_T \mathbf{a}_p &= \mathbf{u}^\parallel - ml \|\boldsymbol{\Omega}\|^2 \mathbf{q} - m_T g \mathbf{e}_z + \mathbf{f}_{a_p}, \\ \dot{\mathbf{q}} &= \boldsymbol{\Omega} \times \mathbf{q}, \\ \dot{\boldsymbol{\Omega}} &= -\frac{1}{ml} \hat{\mathbf{q}}_1 \mathbf{u}^\perp + \mathbf{f}_{a_c}, \\ \dot{\mathbf{R}} &= \mathbf{R} \hat{\boldsymbol{\omega}}, \\ \mathbf{J} \dot{\boldsymbol{\omega}} &= \mathbf{J} \boldsymbol{\omega} \times \boldsymbol{\omega} + \boldsymbol{\tau}_u + \boldsymbol{\tau}_a, \end{aligned} \quad (11)$$

where $m_T = m_p + m$ is the total mass of the payload–quadrotor system. The vectors $\mathbf{p}_p, \mathbf{v}_p, \mathbf{a}_p \in \mathbb{R}^3$ denote the position, velocity, and acceleration of the payload, respectively. The unit vector $\mathbf{q}$ points from the quadrotor position $\mathbf{x} \in \mathbb{R}^3$ toward the payload position $\mathbf{x}_p$, and $\boldsymbol{\Omega} \in \mathbb{R}^3$ represents the angular velocity of the cable.

The total control force applied by the quadrotor is decomposed as

$$\mathbf{u}_0 = \mathbf{u}^\parallel + \mathbf{u}^\perp, \quad (12)$$

where the parallel and perpendicular components are defined as

$$\mathbf{u}^\parallel = \mathbf{q} \mathbf{q}^\top f_u \mathbf{R} \mathbf{e}_z, \quad \mathbf{u}^\perp = -\hat{\mathbf{q}}^2 f_u \mathbf{R} \mathbf{e}_z, \quad (13)$$

and f_u is the collective thrust generated by the quadrotor.

The residual aerodynamic forces acting on the system are represented by $\mathbf{f}_{a_p}$ and $\mathbf{f}_{a_c}$, applied to the payload and cable dynamics, respectively. The last two equations in (11) describe the rotational dynamics of the quadrotor.

Appendix B. Cable-Suspended Payload Geometric-INDI Controller

This section presents the complete mathematical derivation of an alternative geometric controller in [Yu et al. \(2020\)](#) and INDI formulation (our contribution) that explicitly includes the cable dynamics. The purpose is to provide the theoretical foundation complementing the INDI variant used in the main paper.

We adopt a geometric INDI formulation in which the payload tracking is regulated by a virtual force controller $\mathbf{F}_d$ along the cable direction, while the cable attitude is controlled by a perpendicular component. We consider a smooth reference trajectory for the payload, defined by its desired position, velocity, acceleration, and higher-order derivatives up to the n^{th} order $(\mathbf{p}_{p_d}, \mathbf{v}_{p_d}, \mathbf{a}_{p_d}, \mathbf{j}_{p_d}, \mathbf{x}_{p_d}^{(n)})$. Let

$$\mathbf{e}_p = \mathbf{p}_{p_d} - \mathbf{p}_p, \quad \mathbf{e}_v = \mathbf{v}_{p_d} - \mathbf{v}_p, \quad (14)$$

and let the feedback gains be $k_p, k_v > 0$ (scalars) or, more generally, $\mathbf{K}_p, \mathbf{K}_v \in \mathbb{R}^{3 \times 3}$ positive definite (diagonal) matrices. The desired payload virtual force is

$$\mathbf{F}_d = m_T (\mathbf{a}_{p_d} + \mathbf{K}_p \mathbf{e}_p + \mathbf{K}_v \mathbf{e}_v + g \mathbf{e}_z). \quad (15)$$

附录 A. 悬吊电缆载荷动力学

计算机科学

通过将缆绳建模为刚性杆，一个四旋翼单独运输点质量有效载荷的动力学由以下方程决定。根据 Tang 等 (2014)；Sreenath 等 (2013)；Yu 等 (2020)；Li 等 (2023)，耦合四旋翼-载荷动力学可以从哈密顿原理表示为

$$\begin{aligned}
 m_T \mathbf{a}_p &= \mathbf{u}^\parallel - ml \|\boldsymbol{\Omega}\|^2 \mathbf{q} - m_T g \mathbf{e}_z + \mathbf{f}_{a_p}, \\
 \dot{\mathbf{q}} &= \boldsymbol{\Omega} \times \mathbf{q}, \\
 \dot{\boldsymbol{\Omega}} &= -\frac{1}{ml} \hat{\mathbf{q}}_1 \mathbf{u}^\perp + \mathbf{f}_{a_c}, \\
 \dot{\mathbf{R}} &= \mathbf{R} \hat{\boldsymbol{\omega}}, \\
 \mathbf{J} \dot{\boldsymbol{\omega}} &= \mathbf{J} \boldsymbol{\omega} \times \boldsymbol{\omega} + \boldsymbol{\tau}_u + \boldsymbol{\tau}_a,
 \end{aligned} \tag{11}$$

其中 $m_T = m_p + m$ 是有效载荷-四旋翼系统的总质量。向量 $\mathbf{p}_p, \mathbf{v}_p, \mathbf{a}_p \in \mathbb{R}^3$ 分别表示有效载荷的位置、速度和加速度。单位向量 $\mathbf{q}$ 从四旋翼位置 $\mathbf{x} \in \mathbb{R}^3$ 指向有效载荷位置 $\mathbf{x}_p$ ，而 $\boldsymbol{\Omega} \in \mathbb{R}^3$ 表示缆绳的角速度。

四旋翼施加的总控制力被分解为

$$\mathbf{u}_0 = \mathbf{u}^\parallel + \mathbf{u}^\perp, \tag{12}$$

其中平行和垂直分量的定义为

$$\mathbf{u}^\parallel = \mathbf{q} \mathbf{q}^\top f_u \mathbf{R} \mathbf{e}_z, \quad \mathbf{u}^\perp = -\hat{\mathbf{q}}^2 f_u \mathbf{R} \mathbf{e}_z, \tag{13}$$

而 f_u 是由四旋翼产生的总推力。

作用在系统上的剩余空气动力由 $\mathbf{f}_{a_p}$ 和 $\mathbf{f}_{a_c}$ 表示，分别施加在有效载荷和电缆动力学上。(11) 中的最后两个方程描述了四旋翼的旋转动力学。

附录 B. 缆索悬挂载荷几何-INDI 控制器

本节介绍了 Yu 等人 (2020) 中一种替代几何控制器的完整数学推导以及显式包含电缆动力学的 INDI 表述（我们的贡献）。其目的是为主论文中使用的 INDI 变体提供理论基础。

我们采用一种几何 INDI 公式，其中有效载荷跟踪通过沿电缆方向的虚拟力控制器 $\mathbf{F}_d$ 进行调节，而电缆姿态则由垂直分量控制。我们考虑有效载荷的平滑参考轨迹，由其期望位置、速度、加速度以及高阶导数至 n^{th} 阶 ($\mathbf{p}_{p_d}, \mathbf{v}_{p_d}, \mathbf{a}_{p_d}, \mathbf{j}_{p_d}, \mathbf{x}_{p_d}^{(n)}$) 定义。令

$$\mathbf{e}_p = \mathbf{p}_{p_d} - \mathbf{p}_p, \quad \mathbf{e}_v = \mathbf{v}_{p_d} - \mathbf{v}_p, \tag{14}$$

并且让反馈增益为 $k_p, k_v > 0$ (标量) 或更一般地为 $\mathbf{K}_p, \mathbf{K}_v \in \mathbb{R}^{3 \times 3}$ 正定 (对角) 矩阵。期望的有效载荷虚拟力为

$$\mathbf{F}_d = m_T (\mathbf{a}_{p_d} + \mathbf{K}_p \mathbf{e}_p + \mathbf{K}_v \mathbf{e}_v + g \mathbf{e}_z). \tag{15}$$

The parallel control component is then chosen as

$$\mathbf{u}^{\parallel} = \mathbf{q}\mathbf{q}^{\top}\mathbf{F}_d + ml\|\boldsymbol{\Omega}\|^2\mathbf{q} - \mathbf{f}_{a_p}, \quad (16)$$

To regulate the cable direction, define the desired direction

$$\mathbf{q}_d = -\frac{\mathbf{F}_d}{\|\mathbf{F}_d\|}, \quad (17)$$

and let $\boldsymbol{\Omega}_d$ satisfy the kinematics $\boldsymbol{\Omega}_d = \dot{\mathbf{q}}_d \times \mathbf{q}_d$, where $\dot{\mathbf{q}}_d$ is computed with numerical differentiation from $\mathbf{q}_d$ or ideally from the differential flatness computation in [Tang et al. \(2014\)](#) as

$$\dot{\mathbf{q}}_d = \frac{-m_p \mathbf{j}_{p_d} + \dot{T} \mathbf{q}_d}{\|m_p \mathbf{a}_{p_d} + g \mathbf{e}_z\|}, \quad \dot{T} = -m_p (\mathbf{j}_{p_d} \cdot \mathbf{q}_d). \quad (18)$$

Define the cable attitude and angular-rate errors (consistent with [Sreenath et al. \(2013\)](#); [Yu et al. \(2020\)](#)) as

$$\mathbf{e}_q = \mathbf{q}_d \times \mathbf{q}, \quad \mathbf{e}_{\Omega} = \boldsymbol{\Omega} + \hat{\mathbf{q}}^2 \boldsymbol{\Omega}_d, \quad (19)$$

and choose positive gains $k_q, k_{\Omega} > 0$ (or $\mathbf{K}_q, \mathbf{K}_{\Omega}$ positive definite diagonal matrices). Using the cable dynamics in (11), a perpendicular control that stabilizes $(\mathbf{q}, \boldsymbol{\Omega}) \rightarrow (\mathbf{q}_d, \boldsymbol{\Omega}_d)$ is

$$\mathbf{u}^{\perp} = ml \hat{\mathbf{q}}_1 (-\mathbf{K}_q \mathbf{e}_q - \mathbf{K}_{\Omega} \mathbf{e}_{\Omega} - (\mathbf{q} \cdot \boldsymbol{\Omega}_d) \dot{\mathbf{q}} - \hat{\mathbf{q}}^2 \dot{\boldsymbol{\Omega}}_d - \mathbf{f}_{a_c}), \quad (20)$$

Finally, The collective thrust of the quadrotor is defined as $f = \mathbf{u}_0 \cdot \mathbf{R} \mathbf{e}_z$, and the attitude is tracked using the geometric controller proposed in [Lee et al. \(2010\)](#).

B.1. INDI Formulation

In the geometric INDI framework, the residual aerodynamic forces $\mathbf{f}_{a_p}$ and $\mathbf{f}_{a_c}$ correspond to the unmodeled disturbances acting on the payload and the cable, respectively, as defined in the dynamics equations in (11). The term $\mathbf{f}_{a_p}$ captures the discrepancy between the measured acceleration of the payload $\ddot{\mathbf{x}}_p$ (left hand side) and the nominal acceleration predicted by the model (right hand side). Similarly, $\mathbf{f}_{a_c}$ is derived analogously from the measured angular acceleration of the cable $\dot{\boldsymbol{\Omega}}$. These residuals are used by the INDI controller to incrementally adjust the control inputs based on the difference between the measured and model-predicted dynamics as shown in Eqs. (16) and (20).

B.1.1. PRACTICAL IMPLEMENTATION

While the previous section provides the theoretical derivation, the practical realization of this formulation presents significant challenges. We outline below the sensing and numerical differentiation requirements, along with the sources of estimation noise that make the real-time implementation difficult on lightweight multirotors hardware. In our experimental setup, the payload position $\mathbf{p}_p$ is measured directly from the motion capture (MoCap) system. The payload acceleration $\mathbf{a}_p$ is estimated by twice numerically differentiating the measured position data, with a Butterworth filter applied after each differentiation step to obtain smooth velocity and acceleration estimates.

Similarly, the cable direction $\mathbf{q}$ is computed from the measured positions of the quadrotor and the payload. The cable angular velocity $\boldsymbol{\Omega}$ and angular acceleration $\dot{\boldsymbol{\Omega}}$ are estimated through numerical differentiation of $\mathbf{q}$ and filtered using the same Butterworth approach (applied after each

然后选择并联控制组件为

$$\mathbf{u}^{\parallel} = \mathbf{q}\mathbf{q}^{\top}\mathbf{F}_d + ml\|\boldsymbol{\Omega}\|^2\mathbf{q} - \mathbf{f}_{a_p}, \quad (16)$$

要调节电缆方向，先确定所需方向

$$\mathbf{q}_d = -\frac{\mathbf{F}_d}{\|\mathbf{F}_d\|}, \quad (17)$$

并让 $\boldsymbol{\Omega}_d$ 满足运动学 $\boldsymbol{\Omega}_d = \dot{\mathbf{q}}_d \times \mathbf{q}_d$ ，其中 $\dot{\mathbf{q}}_d$ 是通过通过对 $\mathbf{q}_d$ 进行数值微分计算的，或理想情况下通过 Tang 等人 (2014) 的微分平坦性计算得出

$$\dot{\mathbf{q}}_d = \frac{-m_p \mathbf{j}_{p_d} + \dot{T} \mathbf{q}_d}{\|m_p \mathbf{a}_{p_d} + g \mathbf{e}_z\|}, \quad \dot{T} = -m_p (\mathbf{j}_{p_d} \cdot \mathbf{q}_d). \quad (18)$$

将电缆姿态和角速度误差定义为（与 Sreenath 等人 (2013); Yu 等人 (2020) 一致）

$$\mathbf{e}_q = \mathbf{q}_d \times \mathbf{q}, \quad \mathbf{e}_{\Omega} = \boldsymbol{\Omega} + \hat{\mathbf{q}}^2 \boldsymbol{\Omega}_d, \quad (19)$$

并选择正增益 $k_q, k_{\Omega} > 0$ （或 $\mathbf{K}_q, \mathbf{K}_{\Omega}$ 正定对角矩阵）。使用公式 (11) 中的电缆动力学，一个垂直控制来稳定 $(\mathbf{q}, \boldsymbol{\Omega}) \rightarrow (\mathbf{q}_d, \boldsymbol{\Omega}_d)$ 是

$$\mathbf{u}^{\perp} = ml \hat{\mathbf{q}}_1 (-\mathbf{K}_q \mathbf{e}_q - \mathbf{K}_{\Omega} \mathbf{e}_{\Omega} - (\mathbf{q} \cdot \boldsymbol{\Omega}_d) \dot{\mathbf{q}} - \hat{\mathbf{q}}^2 \dot{\boldsymbol{\Omega}}_d - \mathbf{f}_{a_c}), \quad (20)$$

最后，四旋翼的总推力定义为 $f = \mathbf{u}_0 \cdot \mathbf{R} \mathbf{e}_z$ ，姿态则使用 Lee 等人 (2010) 提出的几何控制器进行跟踪。

B.1. INDI 制剂

在几何 INDI 框架中，残余气动力 $\mathbf{f}_{a_p}$ 和 $\mathbf{f}_{a_c}$ 分别对应于作用在有效载荷和钢索上的未建模扰动，如 (11) 式动力学方程中所定义。项 $\mathbf{f}_{a_p}$ 捕捉了有效载荷测量加速度 $\ddot{\mathbf{x}}_p$ （左侧）与模型预测的名义加速度（右侧）之间的差异。类似地， $\mathbf{f}_{a_c}$ 也是从钢索的测量角加速度 $\ddot{\boldsymbol{\Omega}}$ 类比得出。这些残差由 INDI 控制器使用，以根据测量与模型预测动力学之间的差异增量调整控制输入，如 (16) 式和 (20) 式所示。

B.1.1. 实践实施

虽然前一部分提供了理论推导，但这一公式的实际实现存在重大挑战。我们在下文概述了传感和数值微分的要求，以及使实时实现困难的估计噪声来源。在我们的实验设置中，有效载荷位置 $\mathbf{p}_p$ 直接通过动作捕捉（MoCap）系统测量。有效载荷加速度 $\mathbf{a}_p$ 则通过对测量位置数据进行两次数值微分来估算，每次微分步骤后都应用巴特沃斯滤波器，以获得平滑的速度和加速度估计。

同样，电缆方向 $\mathbf{q}$ 是根据四旋翼和有效载荷的测量位置计算的。电缆角速度 $\boldsymbol{\Omega}$ 和角加速度 $\dot{\boldsymbol{\Omega}}$ 通过对 $\mathbf{q}$ 进行数值微分来估算，并使用相同的巴特沃斯方法进行滤波（在每次应用之后）。

step). In practice, this introduces a trade-off between signal noise and measurement delay: a higher filter cutoff frequency yields a noisier but more responsive signal, while a lower cutoff produces smoother estimates at the cost of additional delay.

Note: In principle, one might implement an Extended Kalman Filter (EKF) for the payload and the cable state as in [Sun et al. \(2025\)](#). However as the payload dynamics are tightly coupled with those of the quadrotor and then the estimation leads to large matrix inversions, which are computationally demanding on the STM32 firmware of the Crazyflie platform.

B.1.2. DISCUSSION

Building on the practical implementation discussed above, we now reflect on the limitations and potential directions for improving the INDI formulation. In principle, the proposed INDI formulation was not experimentally validated in this work, as the estimated states were affected either by significant noise introduced through numerical differentiation or by the delays introduced when using low Butterworth filter cut-off frequencies. In particular, the estimation of the cable angular acceleration $\ddot{\Omega}$ proved highly sensitive to measurement noise, even after filtering, which led to residual force estimates that were too noisy to yield reliable experimental results. Nevertheless, we include this formulation as it represents the theoretically consistent approach derived directly from the system dynamics.

In contrast, the INDI formulation used in the main paper represents a trade-off between modeling fidelity and practical feasibility. By dropping the explicit cable dynamics and relying on IMU-measured accelerations of the multirotors (which already capture the coupling effects through the cable tension) the controller assumes perfect knowledge of these effects while gaining smoother residual estimates and improved real-time stability.

A promising direction to address these challenges is to learn the residual dynamics directly using a data-driven model, where temporal input histories implicitly capture higher-order state information, thereby reducing the reliance on explicit state estimation. This analysis provides the foundation for future work on integrating learned residual models and state estimation methods for reliable INDI-based control of cable-suspended systems.

步骤)。在实际操作中，这引入了信号噪声与测量延迟之间的权衡：较高的滤波器截止频率会产生噪声更大但响应更迅速的信号，而较低的截止频率则会以增加延迟为代价生成更平滑的估计。

Note: 原则上，可以像 Sun 等人 (2025) 那样为有效载荷和绳索状态实现扩展卡尔曼滤波器 (EKF)。然而，由于有效载荷的动力学与四旋翼的动力学紧密耦合，因此估计会导致大矩阵求逆，这在 Crazyflie 平台的 STM32 固件上计算量很大。

B.1.2. 讨论

在上述实际实现的基础上，我们现在反思 INDI 公式的局限性以及改进的潜在方向。原则上，本文提出的 INDI 公式并未经过实验验证，因为估计的状态要么受到数值微分引入的显著噪声影响，要么受到使用低巴特沃斯滤波器截止频率时引入的延迟影响。特别是，电缆角加速度 $\ddot{\Omega}$ 的估计对测量噪声非常敏感，即使在滤波之后，这也导致剩余力的估计噪声过大，无法产生可靠的实验结果。尽管如此，我们仍然包括该公式，因为它代表了从系统动力学直接推导出的理论上一致的方法。

相比之下，主要论文中使用的 INDI 公式体现了建模精度与实际可行性之间的权衡。通过放弃显式的绳索动力学，并依赖多旋翼的 IMU 测得的加速度（这些加速度已经通过绳索张力捕获了耦合效应），控制器假设对这些效应有完全的了解，同时获得更平滑的残差估计和改进的实时稳定性。

解决这些挑战的一个有前途的方向是使用数据驱动模型直接学习残差动力学，其中时间输入历史隐式地捕提高阶状态信息，从而减少对显式状态估计的依赖。这一分析为未来将学习的残差模型与状态估计方法结合，以实现基于 INDI 的绳索悬挂系统可靠控制的工作提供了基础。